1. What is the output of the following code?

```
s = "banana"
print(s.count('a'))
```

a) 1

b) 2

c) 3

d) 0

Answer: c) 3

Explanation: `'a'` appears 3 times in `'banana'`.

2. Strings in Python are mutable. (True/False)

a) True

b) False

Answer: b) False

Explanation: Strings are immutable in Python; their contents cannot be changed after creation.

3. What does the following code print?

```
s = "Python"
print(s[1:4])
```

a) Pyt

b) ytho

c) yth

d) ytho

Answer: c) yth

Explanation: `s[1:4]` returns characters at indices 1, 2, and 3.

4. Which string method returns the number of times a substring occurs in the string?

a) find()

b) count()

c) index()

d) replace()

Answer: b) count()

Explanation: `count()` counts occurrences of a substring.

5. What is the output of `"hello".upper()`?

a) HELLO

b) hello

c) Hello

d) hELLO

Answer: a) HELLO

Explanation: `.upper()` converts all characters to uppercase.

6. The expression `s = "abc"; s = "z"` will:

a) Change 'abc' to 'zbc'

b) Raise an error

c) Print 'zbc'

d) Change 'abc' to 'azc'

Answer: b) Raise an error

Explanation: You can’t assign a value to a string character—strings are immutable.

7. What is the result of `"Python".replace("y", "i")`?

a) Pithon

b) Pithy

c) Pithon

d) Pith

Answer: a) Pithon

Explanation: Replaces 'y' with 'i' in `"Python"`.

8. The `.find()` method returns ____ if the substring is not found.

a) True

b) 0

c) -1

d) Error

Answer: c) -1

Explanation: `.find()` returns -1 on failure.

9. Which method is used to check if a string contains only alphabets?

a) isalpha()

b) isalnum()

c) isdigit()

d) isupper()

Answer: a) isalpha()

Explanation: `isalpha()` checks for only alphabetic characters.

10. What will be the value of `s` after this code:

```
s = "  spaces  "
s = s.strip()
```

a) " spaces "

b) "spaces"

c) "spaces "

d) " spaces"

Answer: b) "spaces"

Explanation: `strip()` removes whitespace from both ends.

11. Which method can change "hello world" to "Hello World"?

a) capitalize()

b) title()

c) upper()

d) lower()

Answer: b) title()

Explanation: `title()` capitalizes the first character of every word.

12. What does `s.endswith('a')` return for `s = "Panama"`?

a) True

b) False

Answer: a) True

Explanation: The string ends with 'a'.

13. What is the output of the following?

```
s = "abc"
print(s[5])
```

a) IndexError

b) 'c'

c) None

d) ' '

Answer: a) IndexError

Explanation: The index is out of bounds.

14. What does `"Hello"[::-1]` produce?

a) 'olleH'

b) 'Hello'

c) 'H'

d) ''

Answer: a) 'olleH'

Explanation: `[::-1]` reverses the string.

15. Is `'abc123'.isalnum()` True or False?

a) True

b) False

Answer: a) True

Explanation: Contains only alphabetic and numeric characters.

16. What does `" hello "`.lstrip() return?

a) "hello"

b) " hello"

c) "hello "

d) " hello "

Answer: b) "hello "

Explanation: `lstrip()` removes leading whitespace.

17. What is the output of `"Python3.9"`.isdigit()?

a) True

b) False

Answer: b) False

Explanation: The string contains letters and a dot.

18. What will `":".join(['a', 'b', 'c'])` return?

a) "a🅱️c"

b) "abc"

c) "a,b,c"

d) "a:🅱️:c"

Answer: a) "a🅱️c"

Explanation: `join()` combines with the separator.

19. What does `"hello".capitalize()` do?

a) "Hello"

b) "HELLO"

c) "hello"

d) "Hello "

Answer: a) "Hello"

Explanation: Only the first character is capitalized.

20. The operation `"abc" * 2` results in:

a) Error

b) "abcabc"

c) "aabbcc"

d) None

Answer: b) "abcabc"

Explanation: String repetition.

21. Which statement about strings is incorrect?

a) Strings are immutable

b) Strings can be concatenated

c) Strings can be multiplied by an integer

d) Strings can be indexed with negative numbers

e) None of the above

Answer: e) None of the above

Explanation: All statements are correct.

22. Which slice of `"abcdefg"` gives `"cde"`?

a) [2:5]

b) [3:6]

c) [1:3]

d) [1:5]

Answer: a) [2:5]

Explanation: Characters at positions 2, 3, 4.

23. What is the output?

```
s = "Mississippi"
print(s.replace("i", "I", 2))
```

a) MIssIssippi

b) MIssIssIppI

c) Mssssspp

d) MissIssipI

Answer: a) MIssIssippi

Explanation: Only the first 2 occurrences are replaced.

24. `"foo".zfill(5)` returns:

a) "foo00"

b) "00foo"

c) "foo"

d) "foo000"

Answer: b) "00foo"

Explanation: Pads string with zeros.

25. The result of `"apple".swapcase()` is:

a) Apple

b) aPPLE

c) APPLE

d) apple

Answer: b) aPPLE

Explanation: Switches lower and upper case.

26. What does the code print?

```
x = "abc"
print(chr(ord(x[0])+2))
```

a) a

b) c

c) b

d) c

Answer: b) c

Explanation: ord('a') is 97; 97+2=99 which is 'c'.

27. Which method changes every character to lowercase?

a) lower()

b) upper()

c) title()

d) strip()

Answer: a) lower()

Explanation: Converts string to lowercase.

28. `"python" == "Python"` returns:

a) True

b) False

Answer: b) False

Explanation: String comparison is case-sensitive.

29. `"string".find("ing", 2)` evaluates to:

a) -1

b) 2

c) 3

d) 4

Answer: c) 3

Explanation: 'ing' starts at position 3 in 'string'.

30. Which method removes trailing whitespace?

a) strip()

b) lstrip()

c) rstrip()

d) trim()

Answer: c) rstrip()

Explanation: Removes whitespace on the right side.

31. What does the following print?

```
s = "Test!"
print(s[-2])
```

a) s

b) !

c) e

d) t

Answer: d) t

Explanation: Negative indices count from the end.

32. "hello world".replace('l', '', 1) returns:

a) "helo world"

b) "hello word"

c) "heo world"

d) "hel world"

Answer: d) "hel world"

Explanation: Only the first 'l' is removed.

33. "python".center(9, "*") returns:

a) "***python***"

b) "**python**"

c) "**python*"

d) "***python**"

Answer: c) "**python*"

Explanation: The string (6) is centered in 9 with "*" as fill.

34. Is "HELLO".islower() True or False?

a) True

b) False

Answer: b) False

Explanation: All letters are uppercase.

35. "\tPython".isspace() returns:

a) True

b) False

Answer: b) False

Explanation: There are non-space characters.

36. "1234".isnumeric() returns:

a) True

b) False

Answer: a) True

Explanation: '1234' contains only digits.

37. What will this code print?

```
s = "foo bar"
print(s.partition(" "))
```

a) ('foo', ' ', 'bar')

b) ('foo', 'bar', '')

c) ('foo bar', '', '')

d) ('foo', '', 'bar')

Answer: a) ('foo', ' ', 'bar')

Explanation: `partition()` splits at the first occurrence of the separator.

38. What does "Hi! Hello!!".split('!') produce?

a) ['Hi', ' Hello', '', '']

b) ['Hi', ' Hello', '']

c) ['Hi!', ' Hello!!']
d) ['Hi', ' Hello']

Answer: a) ['Hi', ' Hello', '', '']
Explanation: The separator '!' is split each time; empty strings appear where '!'s are adjacent or at the end.

39. `"Py".isidentifier()` returns:

a) True
b) False

Answer: a) True
Explanation: 'Py' is a valid identifier.

40. `"user@domain.com".endswith('.com')`

a) True
b) False

Answer: a) True
Explanation: The string ends with '.com'.

41. `"A1".isdigit()` returns:

a) True
b) False

Answer: b) False
Explanation: Contains non-digit characters.

42. `"A\nB".splitlines()` returns:

a) ['A', 'B']
b) ['A\nB']
c) ['A', '', 'B']
d) ['A', '\n', 'B']

Answer: a) ['A', 'B']

Explanation: Splits at newline.

43. Which of the following will turn the list ['a','b','c'] to "a,b,c"?

a) ",".join(['a','b','c'])

b) "+".join(['a','b','c'])

c) ['a','b','c'].join(",")

d) str(['a','b','c'])

Answer: a) ",".join(['a','b','c'])

Explanation: The separator is used to join elements.

44. "hello world".split() returns:

a) ['hello', 'world']

b) ['hello', '', 'world']

c) ['hello world']

d) ['h', 'e', 'l', 'l', 'o', 'w', 'o', 'r', 'l', 'd']

Answer: a) ['hello', 'world']

Explanation: Splits by whitespace.

45. "apple".rfind('p') returns:

a) 1

b) 2

c) 3

d) 0

Answer: b) 2

Explanation: Finds the last occurrence.

46. Which of the following is False?

a) "abc".isalpha() is True

b) "".isspace() is True

c) " ".isspace() is True

d) "abc1".isalpha() is False

Answer: b) "".isspace() is True

Explanation: The empty string is False for `isspace()`.

47. "NaN".upper() returns:

a) "nan"

b) "NaN"

c) "NAN"

d) "nAn"

Answer: c) "NAN"

Explanation: Converts all to uppercase.

48. Which produces a syntax error?

a) "hello" + "world"

b) "hello" "world"

c) "hello" - "world"

d) "hello" * 3

Answer: c) "hello" - "world"

Explanation: '-' is not defined for strings.

49. What does ord('A') return?

a) 65

b) 97

c) 66

d) 64

Answer: a) 65

Explanation: Unicode code point of 'A'.

50. 'abcDEF'.islower() returns:

a) True

b) False

Answer: b) False

Explanation: Not all characters are lowercase.

51. What does this print?

```
s = "test"
print(s.capitalize())
```

a) Test

b) test

c) TEST

d) tst

Answer: a) Test

Explanation: Capitalizes only the first letter.

52. Which evaluates to True?

a) " ".isspace()

b) "".isspace()

c) "abc".isspace()

d) " a".isspace()

Answer: a) " ".isspace()

Explanation: All characters are spaces.

53. Which returns True?

```
s = "abcd"
print(s.startswith("ab"))
```

a) True

b) False

Answer: a) True

Explanation: The string starts with 'ab'.

54. What is the result of `"Python".swapcase()`?

a) pYTHON

b) python

c) PYTHON

d) PyThOn

Answer: a) pYTHON

Explanation: Switches cases.

55. Which function converts an integer to a string?

a) str()

b) int()

c) chr()

d) ord()

Answer: a) str()

Explanation: `str()` turns objects into their string representation.

56. "True or False: Python strings can be changed in place."

a) True

b) False

Answer: b) False

Explanation: Strings are immutable.

57. What does `"hi"*3` produce?

a) hi

b) hihihi

c) hi hi hi

d) error

Answer: b) hihihi

Explanation: Repeats 'hi' 3 times.

58. "abc123".isdigit() returns:

a) True

b) False

Answer: b) False

Explanation: There are letters as well as digits.

59. How can you access the last character of a string `s`?

a) s

b) s[-1]

c) s[len(s)]

d) s[len(s)-2]

Answer: b) s[-1]

Explanation: Negative indices start from the end.

60. Which code splits a string into a list of words?

a) s.split()

b) split(s)

c) list(s)

d) s.break()

Answer: a) s.split()

Explanation: The `split()` method breaks the string into words.

61. What does `"42".zfill(5)` produce?

a) '42'

b) '00042'

c) '0420'

d) '42000'

Answer: b) '00042'

Explanation: Pads the string with zeroes on the left.

62. What does `"apple".index('p')` return?

a) 1

b) 2

c) 0

d) Error

Answer: a) 1

Explanation: The first index of 'p' in 'apple' is 1.

63. `"rAnDoM".capitalize()` returns:

a) "Random"

b) "RAnDoM"

c) "rAnDoM"

d) "Randomm"

Answer: a) "Random"

Explanation: Capitalizes first letter, rest becomes lowercase.

64. Is `"" == str()` True or False?

a) True

b) False

Answer: a) True

Explanation: `str()` with no arguments returns an empty string.

65. Which method pads the left side of a string with zeros?

a) pad()

b) lpad()

c) zfill()

d) fill()

Answer: c) zfill()

Explanation: `zfill()` pads with zeros on the left.

66. Which is a valid lambda function that returns the cube of x?

- A. lambda x: x ** 3

- B. lambda x {return x**3}
- C. lambda(x): x**3
- D. lambda x: return x**3

  Answer: A

  Explanation: The correct syntax is `lambda parameters: expression`.

  Difficulty: easy

67. What does the following code output?

```
f = lambda x, y=2: x * y
print(f(3))
```

- A. 2
- B. 3
- C. 6
- D. Error

  Answer: C

  Explanation: Default y=2; so 3 * 2 = 6.

  Difficulty: easy

68. Lambda functions can have multiple expressions.

- A. True
- B. False

  Answer: B

  Explanation: Lambdas are limited to a single expression only.

  Difficulty: easy

69. What is the output?

```
l = [lambda x: x**i for i in range(3)]

print([f(2) for f in l])
```

- **A. [1, 2, 4]**
- **B. [4, 4, 4]**

- **C. [2, 4, 8]**
- **D. Error**

  Answer: B

  Explanation: This is a classic case of **late binding in closures**. The variable `i` is looked up when the lambda is called, not when it's defined. So by the time `f(2)` is called for each lambda, `i` is already `2` for all of them. So: `2**2 = 4` for all.

  Difficulty: hard

70. Which list comprehension returns only even numbers from the range 5?

- A. [x for x in range(5) if x % 2 == 0]
- B. [x for x in range(5) if x % 2]
- C. [x % 2 for x in range(5)]
- D. [x if x%2==0 for x in range(5)]

  Answer: A

  Explanation: Filters even numbers correctly.

  Difficulty: easy

71. List comprehensions can be used to create dictionaries in Python.

- A. True
- B. False

  Answer: A

  Explanation: With curly braces and key:value, e.g. `{k: v for ...}`.

  Difficulty: medium

72. What is a valid way to write a nested list comprehension for a 3×3 matrix of zeros?

- A. [*3]*3
- B. [0 for i in range(3) for j in range(3)]
- C. [[0 for j in range(3)] for i in range(3)]
- D. ,,

  Answer: C

Explanation: Outer list for rows, inner for columns.

Difficulty: medium

73. Given `nums = [1, 2, 3]`, what will `[n*2 for n in nums]` produce?

- **A.** [2, 4, 6]
- **B.** [1, 2, 3, 1, 2, 3]
- **C.** [[1, 1], [2, 2], [3, 3]]
- **D.** [n*2 for n in nums]

Answer: A

Explanation: List comprehension `[n*2 for n in nums]` multiplies each element in `nums` by 2, resulting in `[2, 4, 6]`.

Difficulty: easy

74. Code: Is this list comprehension valid or has an error?

```
[x**2 for x in range(5)]
```

- A. Valid
- B. SyntaxError
- C. IndentationError
- D. TypeError

Answer: A

Explanation: This is the correct syntax for a list comprehension.

Difficulty: easy

75. What will be the following output?

```
a = [1, 2, 3]

b = a

b[0] = 50

print(a)
```

- **A. [1, 2, 3]**
- **B. [50, 2, 3]**

- **C. Error**
- **D. [1, 50, 3]**

  Answer: B

  Explanation: `b = a` **does not create a new list; it creates a new reference to the same list. Changing** `b` **also changes** `a`**. So** `a[0]` **becomes** `50`**.**

  Difficulty: medium

76. Which function creates a shallow copy of list x?

- A. x.copy()
- B. copy.deepcopy(x)
- C. copy(x)
- D. x.clone()

  Answer: A

  Explanation: .copy() creates a shallow copy of a list.

  Difficulty: medium

77. What does deep copy do?

- A. Only copies the outer list
- B. Copies the object and all nested objects
- C. Makes no copy
- D. Deletes all object references

  Answer: B

  Explanation: Deep copy creates a copy of everything recursively.

  Difficulty: easy

78. What will happen here?

```
import copy
matrix = [[1,2],[3,4]]
matrix2 = copy.copy(matrix)
matrix2[0][0] = 99
print(matrix[0][0])
```

- A. 1
- B. 99
- C. Error
- D. 2
  Answer: B
  Explanation: Shallow copy, so nested lists are still shared references.
  Difficulty: medium

79. Code: Is there an error in this code?

```
import copy
a = [1, 2, [3, 4]]
b = copy.deepcopy(a)
b[2][0] = 100
print(a)
```

- A. Correct
- B. TypeError
- C. Output will be [1,2,]
- D. NameError
  Answer: A
  Explanation: Deepcopy ensures a and b are completely independent, so a is unchanged.
  Difficulty: medium

80. Which gives the number of unique values in a list `lst`?

- A. len(lst)
- B. len(set(lst))
- C. lst.count()
- D. sum(lst)
  Answer: B
  Explanation: `set()` removes duplicates, `len()` counts unique items.
  Difficulty: easy

81. What is the output?

```
s = {1, 1, 2, 3}
print(len(s))
```

- A. 3
- B. 4
- C. 2
- D. Error

  Answer: A

  Explanation: Sets cannot have duplicates, only unique elements remain.

  Difficulty: easy

82. What is the correct syntax to add an element '5' to set `s`?

- A. s.add(5)
- B. s.append(5)
- C. add(s, 5)
- D. s = True

  Answer: A

  Explanation: `add()` method is used for sets.

  Difficulty: easy

83. Which option shows a correct tuple declaration?

- A. t = tuple[]
- B. t = (1, 2, 3)
- C. t =
- D. t = tuple(1, 2, 3)

  Answer: B

  Explanation: Rounded brackets denote a tuple.

  Difficulty: easy

84. Tuples in Python are mutable.

- A. True
- B. False

  Answer: B

  Explanation: They are immutable.

  Difficulty: easy

85. Which code will cause an error?

```
d = {'a':1, 'b':2}
d['c'] = 3
print(d)
```

- A. Error
- B. No error

  Answer: B

  Explanation: Adding a new key is valid.

  Difficulty: easy

86. A dict's keys must be:

- A. Mutable
- B. Immutable
- C. Duplicates
- D. Callable

  Answer: B

  Explanation: Keys must be immutable.

  Difficulty: easy

87. What is the output of the following?

```
d = {'a':1, 'b':2}
print('a' in d)
```

- A. False

- B. True

  Answer: B

  Explanation: 'a' is a key in the dictionary.

  Difficulty: easy

88. Which command gives all attributes and methods of object `myobj`?

- A. dir(myobj)
- B. myobj.dir()
- C. listdir(myobj)
- D. attributes(myobj)

  Answer: A

  Explanation: dir() lists all attributes/methods.

  Difficulty: easy

89. Code: Is this statement valid?

```
my_tuple = (1)
```

- A. Valid tuple
- B. Not a tuple
- C. SyntaxError
- D. TypeError

  Answer: B

  Explanation: Parentheses around one value is not a tuple; need comma: `(1,)`

  Difficulty: easy

90. What does `list('python')` return?

- A. ['python']
- B. ['p','y','t','h','o','n']
- C. ['p y t h o n']
- D. Error

  Answer: B

Explanation: Converts each char to list element.

Difficulty: easy

91. Identify the error:

```
s = {1,2,3,4}
s[0]
```

- A. No error
- B. KeyError
- C. TypeError
- D. IndexError

  Answer: C

  Explanation: Sets do not support indexing.

  Difficulty: medium

92. Code: Which line will remove key 'x' from dict d safely?

- A. del d['x']
- B. d.pop('x', None)
- C. d.remove('x')
- D. d.delete('x')

  Answer: B

  Explanation: .pop() with a default avoids a KeyError if the key doesn’t exist.

  Difficulty: medium

93. List comprehensions can create a set. Which syntax is correct?

- A. set([x for x in range(5)])
- B. {x for x in range(5)}
- C. set{x for x in range(5)}
- D. Both A and B

  Answer: D

Explanation: Both create a set from comprehensions.

Difficulty: medium

94. Which code is invalid?

- A. t = (1,2,3)
- B. t = tuple()
- C. t = (1,)
- D. t = (1;2;3)

  Answer: D

  Explanation: Use commas, not semicolons for separating tuple elements.

  Difficulty: easy

95. What will be output?

```
d = {1:"a", 2:"b"}
print(d.get(3, "missing"))
```

- A. 3
- B. missing
- C. Error
- D. None

  Answer: B

  Explanation: .get() returns default if key does not exist.

  Difficulty: easy

96. True or False: The expression `sorted({3,2,1})` produces a list as output.

- A. True
- B. False

  Answer: A

  Explanation: sorted() always returns a list, not a set.

  Difficulty: easy

97. Which command creates an empty dictionary?

- A. dict()
- B. {}
- C. Both
- D. None
  Answer: C
  Explanation: Both are valid ways.
  Difficulty: easy

98. Identify the error in the code:

```
d = dict([('a',1), ('b',2)])
d.add(('c',3))
```

- A. No error
- B. AttributeError
- C. KeyError
- D. SyntaxError
  Answer: B
  Explanation: Dictionaries don’t have an add() method.
  Difficulty: easy

99. What is the output?

```
t = (1, 2, 3)
print(t[1])
```

- A. 0
- B. 1
- C. 2
- D. Error
  Answer: C
  Explanation: Tuples support indexing: t is the second item.
  Difficulty: easy

100. Code: Is this syntactically valid?

```
{ x:y for x, y in zip([1,2], ['a','b']) }
```

- A. Valid
- B. SyntaxError
- C. TypeError
- D. NameError

  Answer: A

  Explanation: This is a dictionary comprehension.

  Difficulty: medium